Literature Review - Best Text-to-Video Generation Model

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Scope and selection criteria

I surveyed open/open-weight T2V models released from 2024-2026. Models were evaluated for availability of weights/code, local inference feasibility on <= 32GB system memory/consumer GPUs (such as an RTX 3080), video quality, temporal consistency, human subject consistency, prompt adherence, resolution, and ability to fine-tune or modify the architecture.

Search methodology

To identify relevant models, my search spanned academic and open-source repositories, including arXiv, Google Scholar, the Hugging Face Hub, and official developer technical reports/GitHub repositories. Search query families were organized into three main tracks: architectural discovery (e.g., "open-weight text-to-video DiT", "open source video foundation model 2025 2026"), hardware and deployment feasibility (e.g., "T2V minimum VRAM FP8 quantization", "consumer GPU video generation benchmark"), and qualitative community evaluation (e.g., "temporal consistency human anatomy video generation reviews", "prompt adherence attribute bleeding comparison").

My exclusion criteria consisted of proprietary, API-only, or closed-source commercial models (e.g., Runway Gen-3, Sora, Kling, Luma Dream Machine) as well as T2V generation capability, minimum local inference requirements exceeding a consumer workstation baseline (e.g., no more than 32GB VRAM), and active releases between 2024-2026.

Comparison table

Open weight model search criteria

The above spreadsheet includes publications for each listed model. Star ratings are based on an aggregate of sentiment rather than benchmark scores or actual testing. I used forums such as Reddit and HackerNews to gauge ……. One row corresponds to one model.

Short synthesis

In my opinion, LTX-2 (Lightricks) and HunyuanVideo-1.5 emerged as the most practical candidates for initial testing because their documented memory requirements fit consumer hardware, while Wan2.2-TI2V-5B and LTX-2.5 (Lightricks) offer stronger higher-capacity alternatives if sufficient GPU memory is available.

Recommended testing method

Question: Which model most reliably produces usable pairs of tattoo images from the same synthetic person under our hardware constraints?

1. Create a small fixed prompt set of maybe 10-20 structured prompts using our schema with varying tattoo location, design, skin tone, gender/presentation, camera angle, clothing, and scene. From these structured prompts, we build full prompt strings that are standardized so that every model gets as similar a task as possible.
2. Run our shortlisted models (see Short synthesis) on exactly the same prompts. Keep resolution, video duration, number of inference steps, guidance scale, and seed as comparable as the models allow.
3. Generate multiple samples per prompt (maybe 2-3), same seed for every prompt
4. Evaluate each video w/ a fixed rubrics, maybe with scores for:
    a. Human realism/anatomy
    b. Tattoo visual quality
    c. Tattoo consistency across frames
    d. Person/identity consistency across frames
    e. Prompt adherence
    f. Temporal coherence
    g. Number of usable frame pairs
5. All 3 of us should score outputs independently, then we can average the scores. The project description explicitly asks for subjective human evaluation w/ multiple annotators.
6. Record practical performance, such as peak VRAM, system RAM, generation time, crashes/OOMs, setup difficulty, and whether fine-tuning is practical.
7. Extract the best frame pair from each video (we should have a minimum of 10 prompts * 2 samples * 3 models = 60 videos). An example quantitative metric we could use might look something like:
    a. Usable pair rate = # of generations production at least one valid tattoo-image pair / total generations